**D 50660** **(Pages : 2)** **Name.....**

**Reg. No.....**

**FIFTH SEMESTER (CBCSS–UG) DEGREE EXAMINATION**
**NOVEMBER 2023**

Mathematics

MTS 5B 09—INTRODUCTION TO GEOMETRY

(2019 Admission only)

Time : Two Hours Maximum : 60 Marks

**Section A**

*Answer any number of questions.*
*Each question carries* 2 *marks.*
*Ceiling is* 20.

1. Let $x=\frac{1}{2}t^2, y=t, t\in\mathbb{R}$ be the parametric equation of a parabola E. Determine the equation of the chord that joins distinct points P and Q on E with parameters $t_1$ and $t_2$ respectively. Determine the condition on $t_1$ and $t_2$ such that the chord PQ passes through the focus of E.

2. Let $x=2\cos t+\cos 2t+1, y=2\sin t+\sin 2t$ be a parametric equation of a curve in $\mathbb{R}^2$, where $t$ is not a multiple of $\pi$. Find the equation of the tangent to this curve at $t=\frac{\pi}{3}$.

3. Let $x=t^2, y=2t, t\in\mathbb{R}$ be parametric equation of the parabola. Find the equation of the normal to the parabola at P with parameter $t$.

4. Classify the non-degenerate conic with equation $3x^2-8xy+2y^2-2x+4y-16=0$.

5. Prove that $\begin{bmatrix}\cos\theta & \sin\theta\\ \sin\theta & -\cos\theta\end{bmatrix}$ is orthogonal for each real number $\theta$.

6. Let $t_1(\mathrm{X})=\begin{pmatrix}1 & 3\\ 1 & 2\end{pmatrix}\mathrm{X}+\begin{pmatrix}4\\ -2\end{pmatrix}$ and $t_2(\mathrm{X})=\begin{pmatrix}-6 & 5\\ 3 & 2\end{pmatrix}\mathrm{X}+\begin{bmatrix}2\\ 1\end{bmatrix}$ be two affine transformations. Determine $t_1\circ t_2$ and state whether the composition is an affine transformation.

7. Find the inverse of the affine transformation $t(\mathrm{X})=\begin{bmatrix}1 & 3\\ 1 & 2\end{bmatrix}\mathrm{X}+\begin{pmatrix}4\\ -2\end{pmatrix}$.

**Turn over**

8. Determine the affine transformation which maps the points (0, 0), (1, 0) and (0, 1) to the points (2, 3), (1, 6) and (3, –1) respectively.
9. State Menelau's theorem.
10. Prove that every hyperbola is affine congruent to the rectangular hyperbola $xy = 1$.
11. Determine whether the points [2, 1, 3], [1, 2, 1] and [–1, 4, –3] are collinear.
12. Determine the point of intersection of the lines $x - y - z = 0$ and $x + 5y + 2z = 0$.

(Ceiling 20)

## Section B

*Answer any number of questions.*
*Each question carries* 5 *marks.*
*Ceiling is* 30.

13. Derive the standard form of the equation of the ellipse.
14. State and prove reflection property of the parabola.
15. Prove that a perpendicular from a focus of an ellipse to a tangent meets the tangent on the auxiliary circle of the ellipse.
16. Prove that a $2 \times 2$ matrix P represents a rotation of $\mathbb{R}^2$ about the origin if and only if P is orthogonal and det P = 1.
17. Prove that the set of all affine transformations A(2) forms a group under the operation of composition of functions.
18. Determine the image of the line $y = 2x$ under the affine transformation $t(X) = \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix} X + \begin{pmatrix} 2 \\ -1 \end{pmatrix}$, $X \in \mathbb{R}^2$.
19. State and prove fundamental theorem of affine geometry.

(Ceiling 30)

## Section C

*Answer any* **one** *question.*
*The question carries* 10 *marks.*
*Maximum marks* 10.

20. Classify the conic $x^2 - 4xy + 4y^2 - 6x - 8y + 5 = 0$ in $\mathbb{R}^2$ and also determine its centre and axis.
21. State and prove Ceva's theorem.

(1 × 10 = 10 marks)